

✓ Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset

Introduction

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset. The analysis uses Python libraries such as Pandas, Matplotlib, and Seaborn to explore the dataset, identify patterns, trends, relationships, and potential anomalies.

The analysis includes statistical exploration, data visualization, missing value analysis, and correlation analysis to gain meaningful insights from the Titanic dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set the style for visualizations
sns.set_style("whitegrid")
```

```
from google.colab import files
uploaded = files.upload()
```

train.csv

train.csv(text/csv) - 61194 bytes, last modified: 9/4/2026 - 100% done
Saving train.csv to train (1).csv

```
df = pd.read_csv("train.csv")

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
				Heikkinen,					STON/O2

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(891, 12)
```

```
df.tail()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp       891 non-null    int64
7   Parch       891 non-null    int64
8   Ticket      891 non-null    object
9   Fare        891 non-null    float64
```

```
10 Cabin      204 non-null object
11 Embarked   889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

```
df.shape
```

```
(891, 12)
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

df.duplicated().sum()

np.int64(0)

df["Survived"].value_counts()

	count
Survived	
0	549
1	342

dtype: int64

df["Cabin"].value_counts()

	count
Cabin	
G6	4
C23 C25 C27	4
B96 B98	4
F2	3
D	3
...	...
E17	1
A24	1
C50	1
B42	1
C148	1

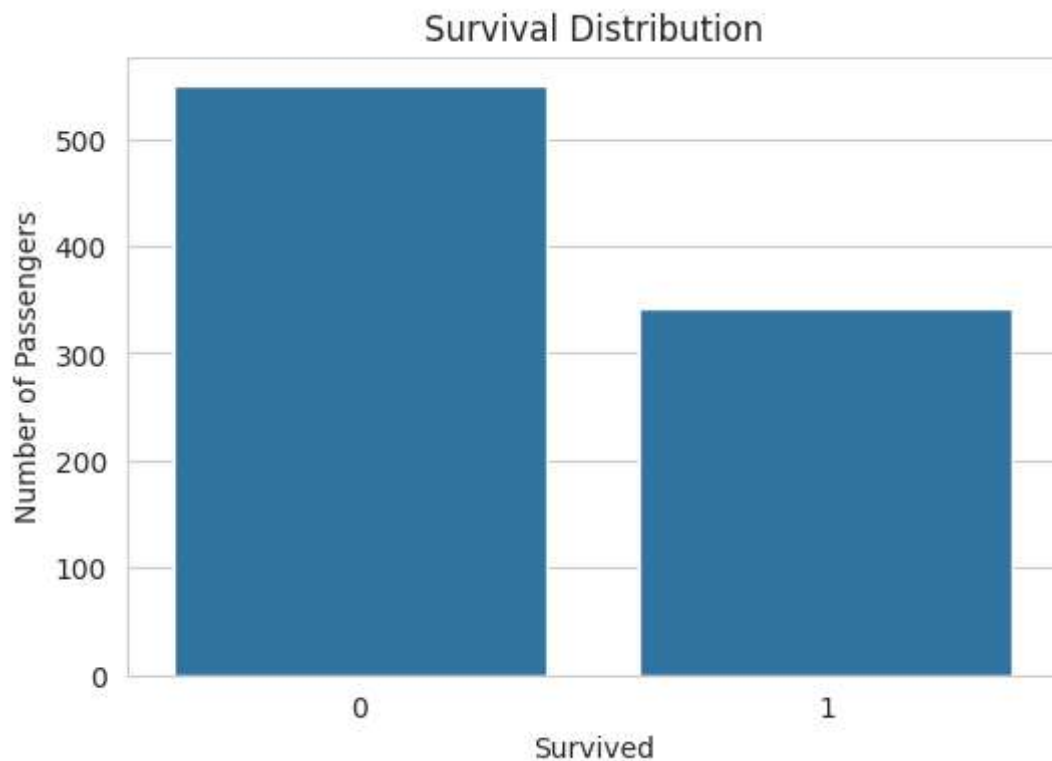
147 rows × 1 columns

dtype: int64

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Survived")

plt.title("Survival Distribution")
plt.xlabel("Survived")
plt.ylabel("Number of Passengers")

plt.show()
```



✓ Observation

The visualization shows the distribution of passengers who survived and those who did not survive. The number of passengers who did not survive was higher than the number of passengers who survived.

```
df["Sex"].value_counts()
```

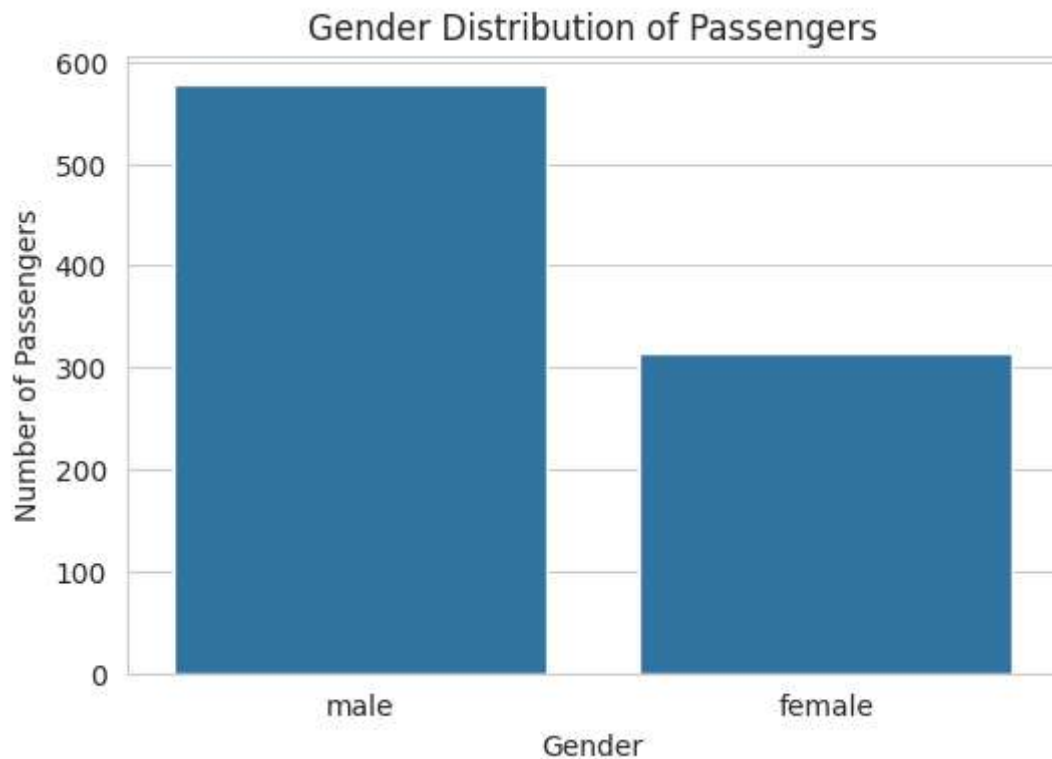
	count
male	577
female	314

dtype: int64

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex")

plt.title("Gender Distribution of Passengers")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```



✓ Observation

The dataset contains more male passengers than female passengers, showing an unequal gender distribution among the recorded passengers.

```
df["Pclass"].value_counts()
```

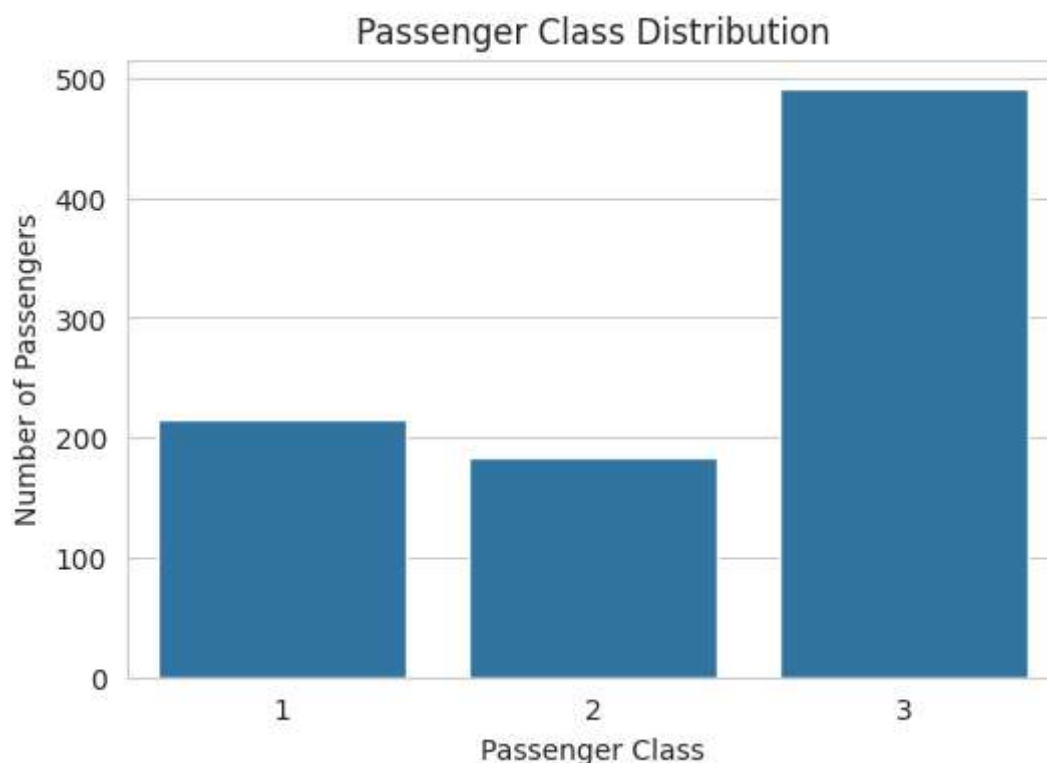
	count
Pclass	
3	491
1	216
2	184

dtype: int64

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Pclass")

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



✓ Observation

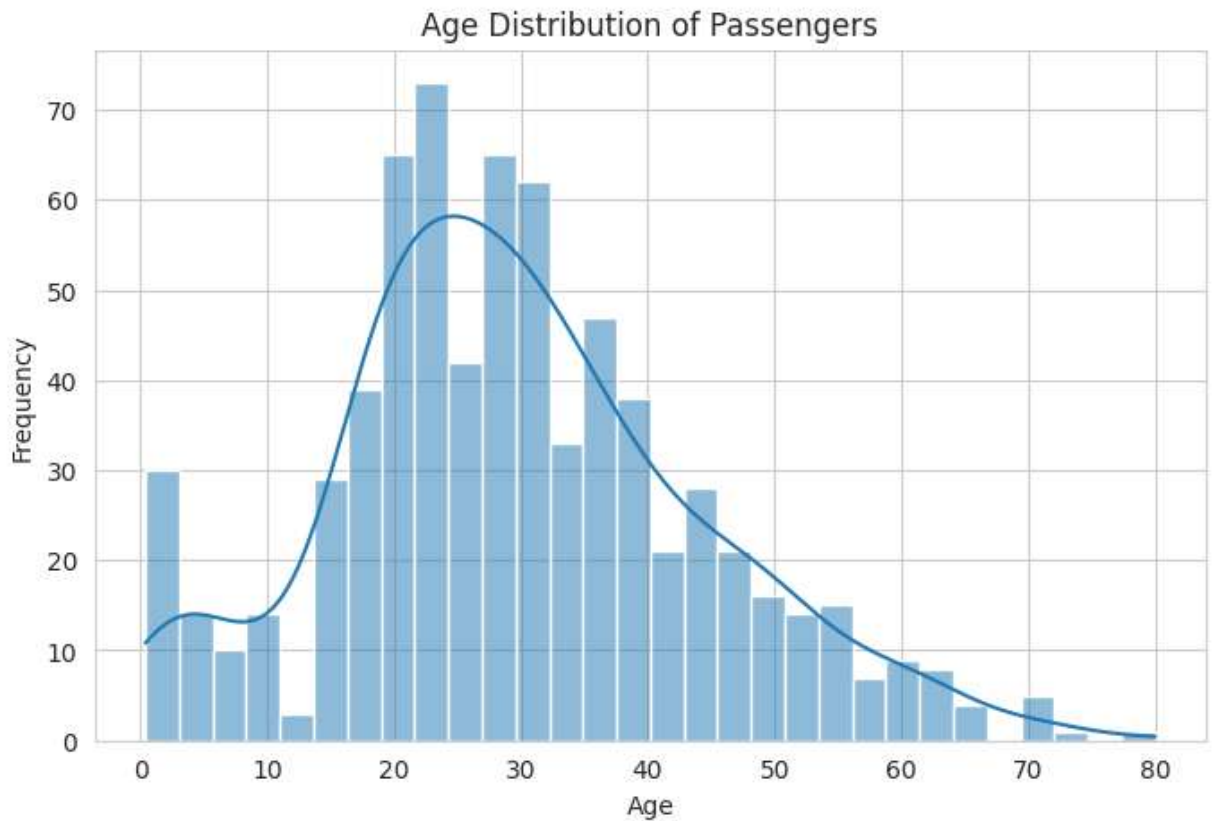
Passengers were distributed across three different ticket classes. The distribution helps identify how many passengers travelled in each class.

```
plt.figure(figsize=(8, 5))

sns.histplot(df["Age"].dropna(), bins=30, kde=True)

plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Frequency")

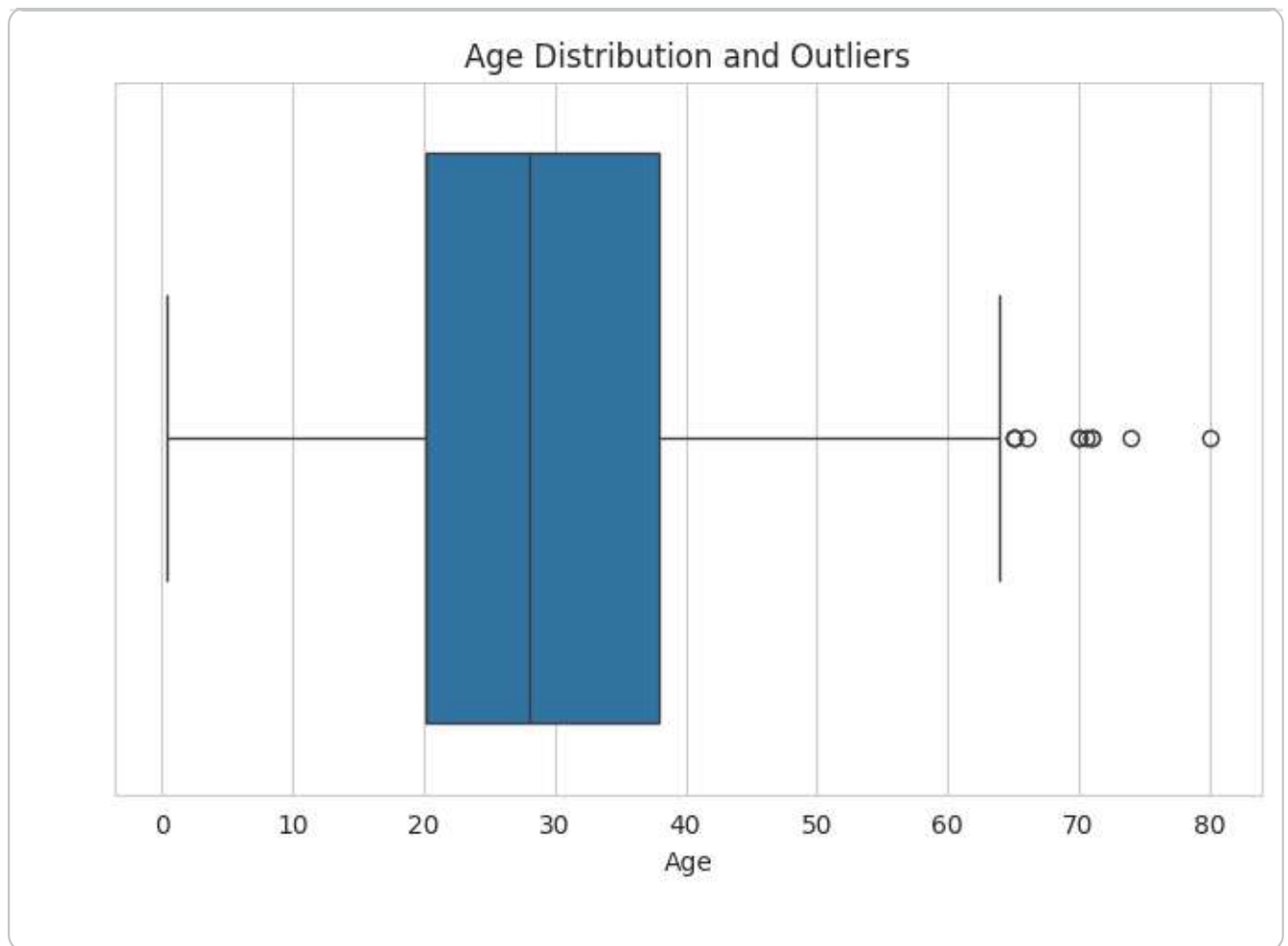
plt.show()
```

✓ Observation

The age distribution shows that passengers belonged to different age groups, with a large proportion of passengers being young adults. The distribution also helps identify the overall spread of passenger ages.

```
plt.figure(figsize=(8, 5))  
  
sns.boxplot(x=df["Age"])  
  
plt.title("Age Distribution and Outliers")  
  
plt.show()
```



✓ Observation

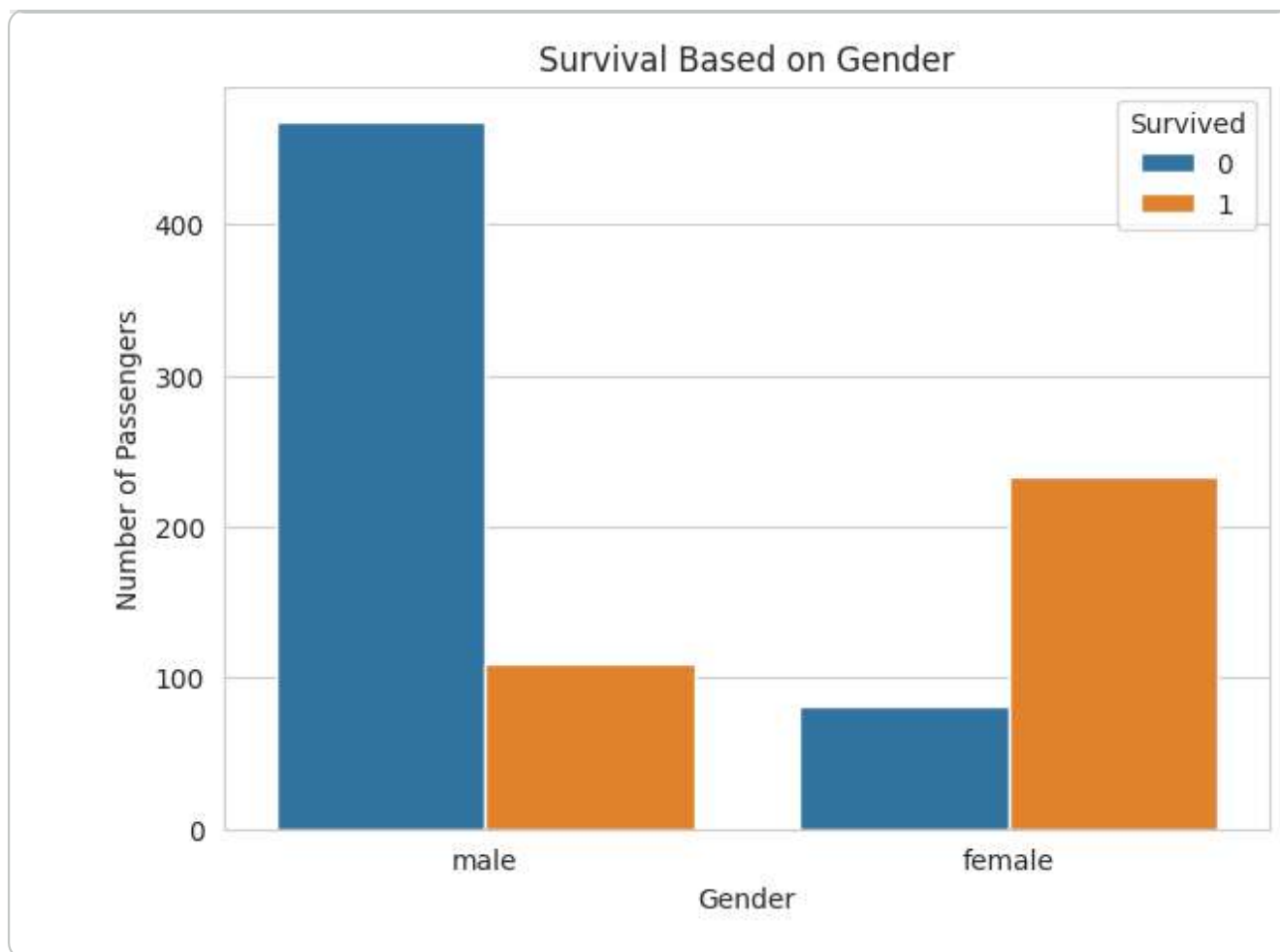
The boxplot displays the median age and overall spread of passenger ages. It can also help identify potential outliers in the age variable.

```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival Based on Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```



✓ Observation

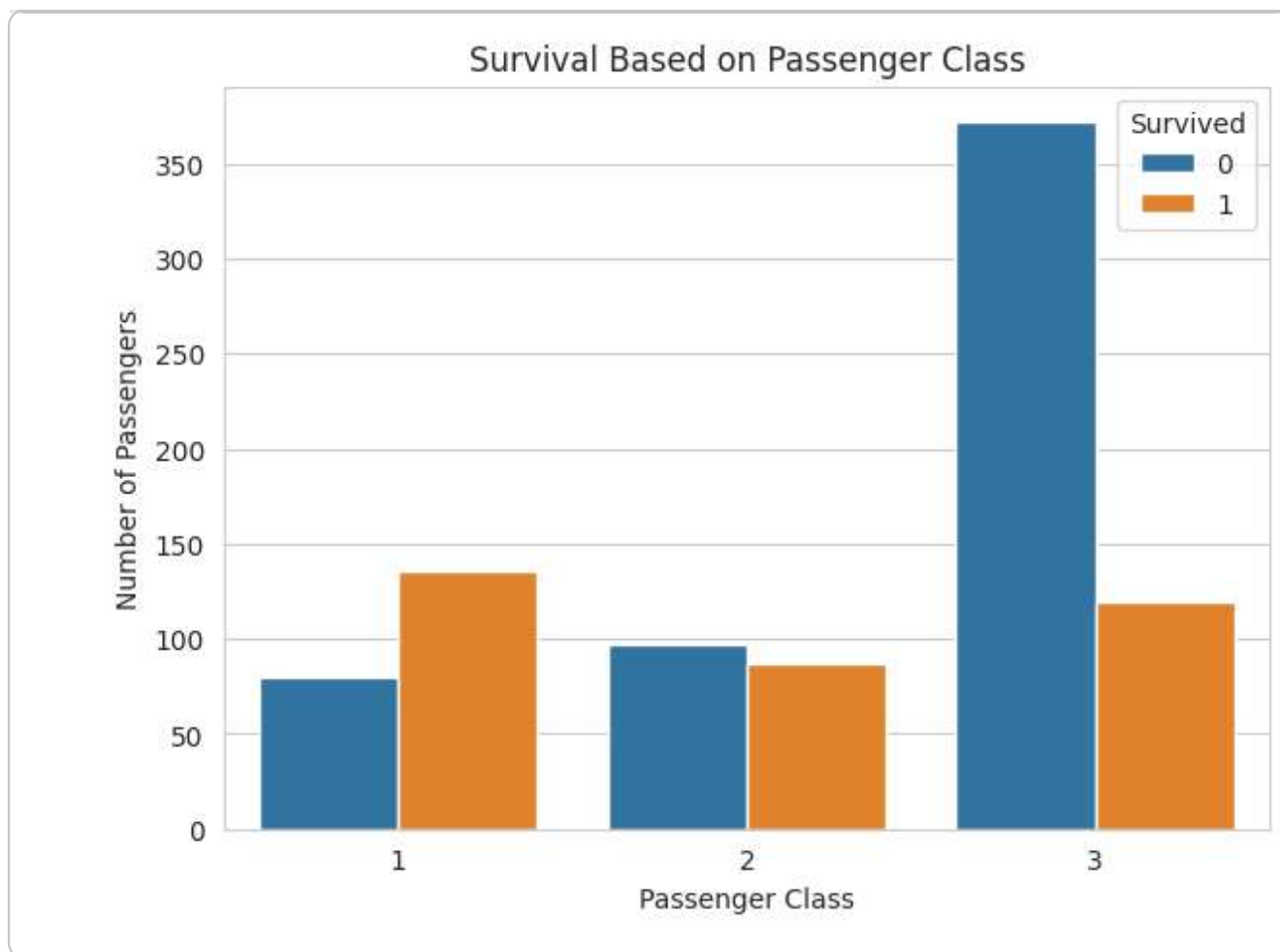
Female passengers had a higher survival rate compared to male passengers. This indicates that gender was an important factor associated with survival.

```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Pclass", hue="Survived")

plt.title("Survival Based on Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



✓ Observation

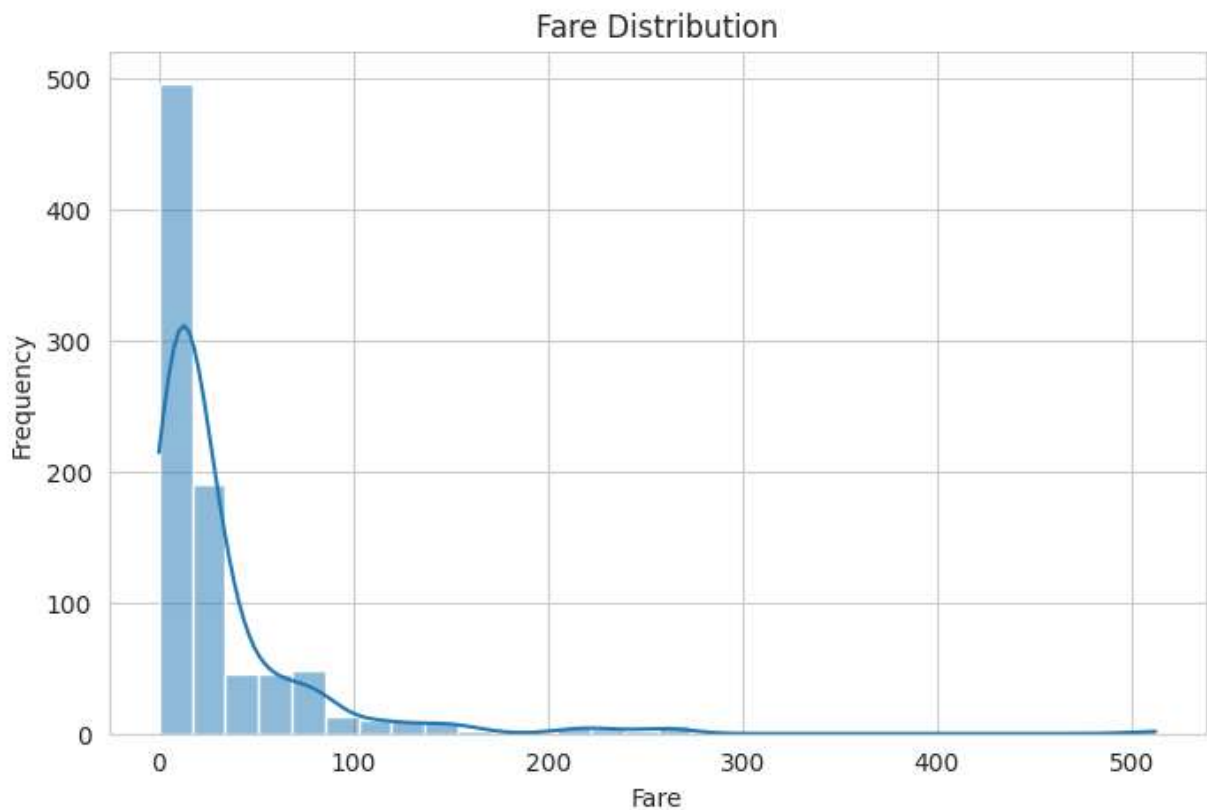
Passenger class showed a noticeable relationship with survival. Passengers in higher classes generally had better survival outcomes compared with passengers in lower classes.

```
plt.figure(figsize=(8, 5))

sns.histplot(df["Fare"], bins=30, kde=True)

plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")

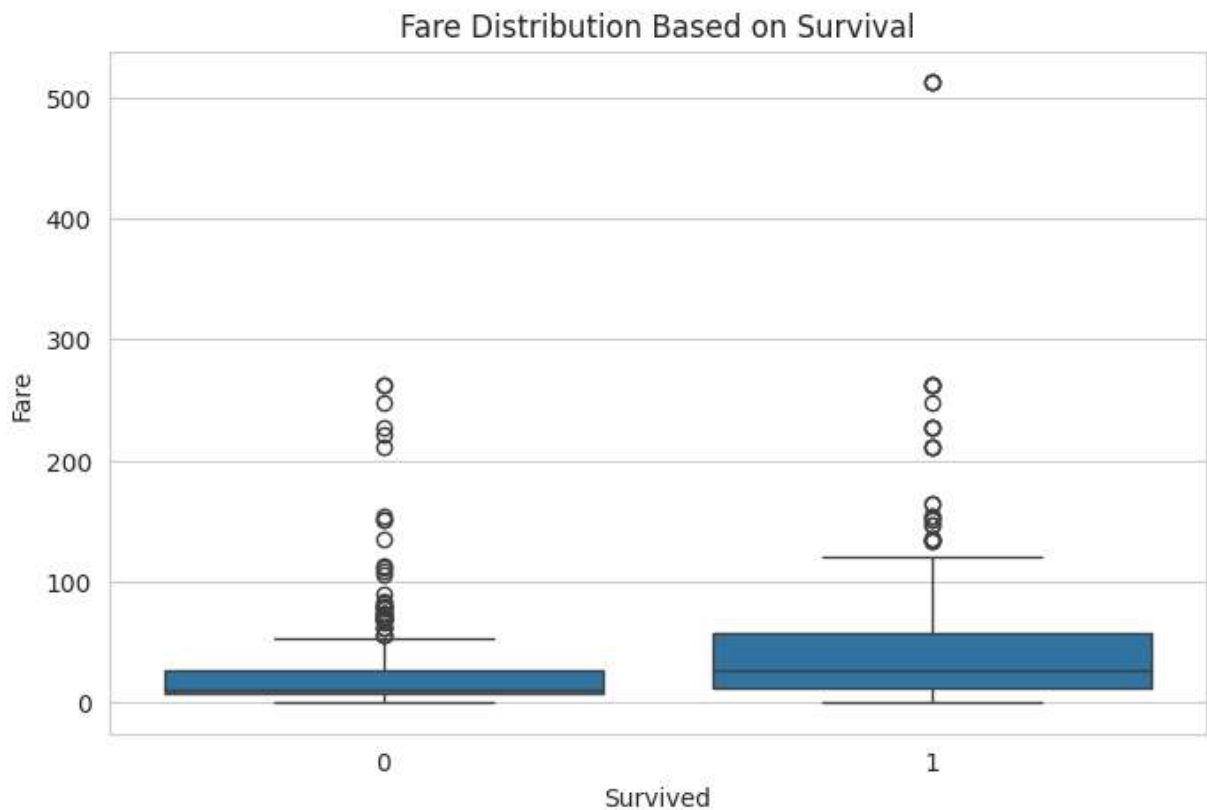
plt.show()
```



✓ Observation

The fare distribution is positively skewed, with most passengers paying lower or moderate fares and a smaller number of passengers paying significantly higher fares.

```
plt.figure(figsize=(8, 5))  
  
sns.boxplot(data=df, x="Survived", y="Fare")  
  
plt.title("Fare Distribution Based on Survival")  
  
plt.show()
```



✓ Observation

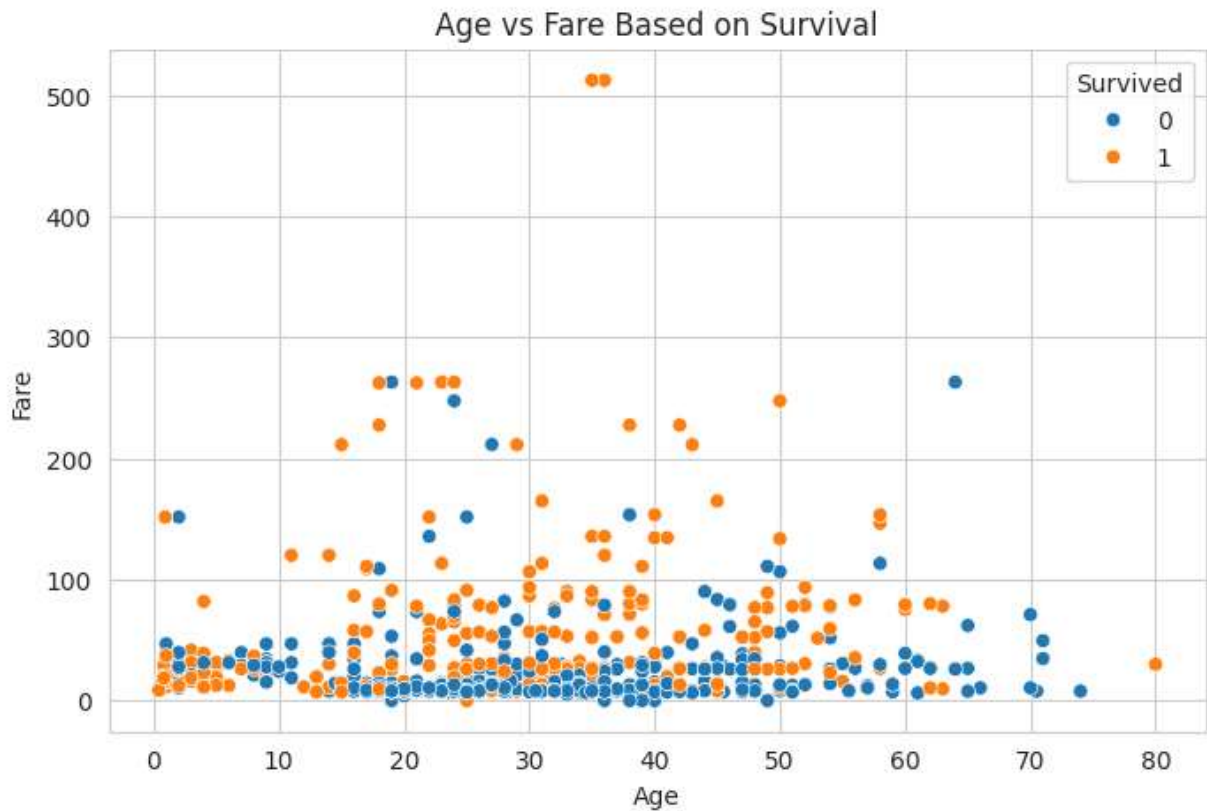
The fare distribution differs between passengers who survived and those who did not survive. This suggests that fare and passenger survival may have some relationship.

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Age",
    y="Fare",
    hue="Survived"
)

plt.title("Age vs Fare Based on Survival")

plt.show()
```

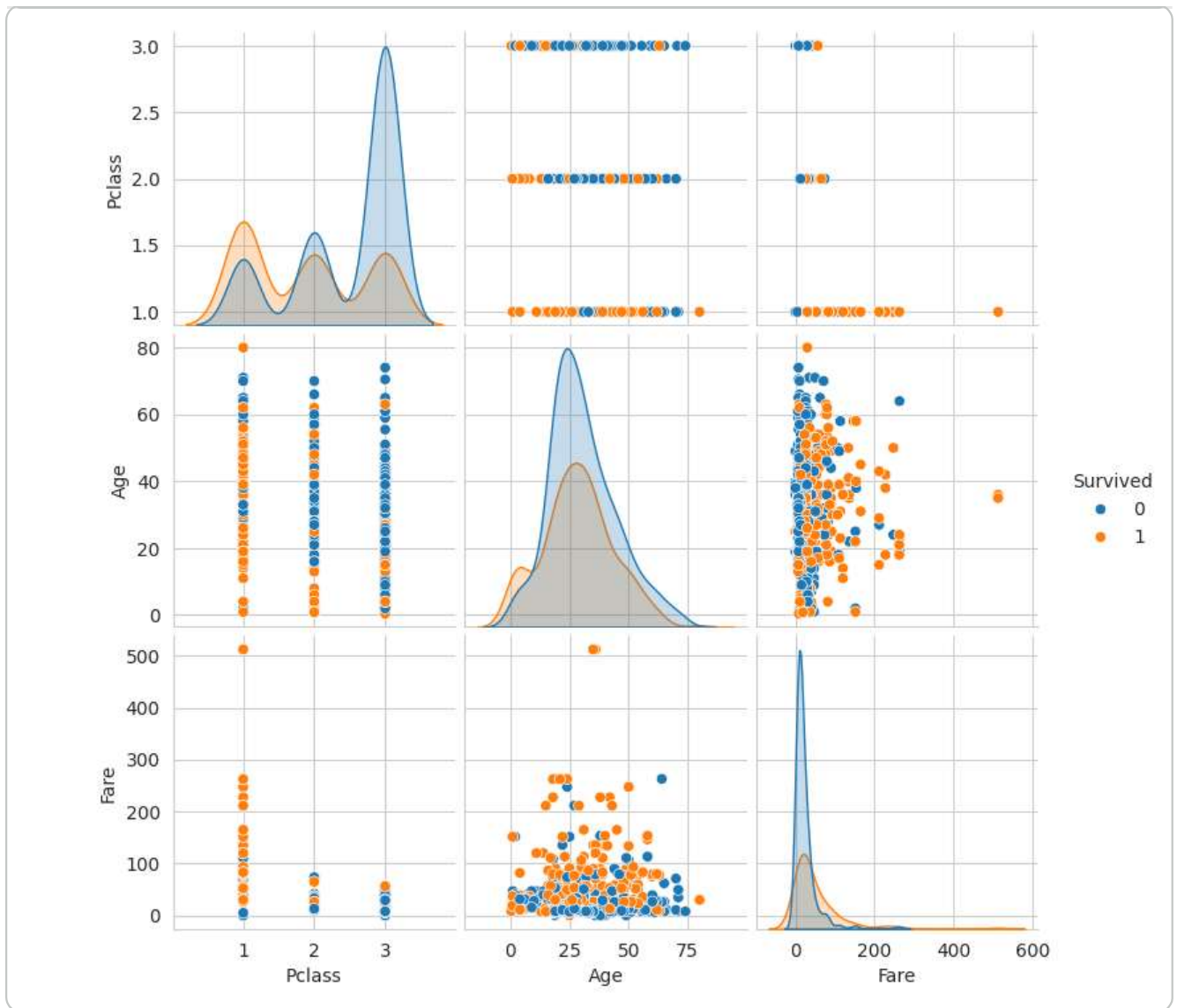


✓ Observation

The scatterplot shows the relationship between passenger age and ticket fare while also displaying survival outcomes. The data points reveal variations in fares across different age groups.

```
sns.pairplot(
    df[["Survived", "Pclass", "Age", "Fare"]].dropna(),
    hue="Survived"
)

plt.show()
```



✓ Observation

The pairplot provides a comparison of multiple numerical variables and helps identify patterns, distributions, and possible relationships between them.